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From speaking up to shutting down: The cognitive costs of non-instrumental voice

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ABSTRACT

In hierarchical organizations, repeated experiences of unacknowledged voice can lead to acquiescent silence—a passive withdrawal rooted in perceived futility. While its emotional and attitudinal consequences are well documented, less is known about its cognitive costs. This study investigates whether repeated voice non-endorsement impairs executive functioning through mechanisms consistent with the informational learned helplessness framework. In a controlled laboratory experiment ($N = 129$), participants were exposed to either low, high, or no feedback regarding the impact of their suggestions in a simulated consultation task. Subsequently, they completed the Parametric Go/No-Go task, a validated measure of executive performance assessing sustained attention, inhibition, shifting, updating, and processing speed across three levels involving distinct types of executive demands. Participants in the low voice impact condition showed significantly reduced performance on key executive indices (particularly inhibitory control and efficiency) compared to the high-impact and control groups, exclusively at Level 2 of the task, where adaptive strategy use is required. These impairments were accompanied by lower self-reported motivation and perceived control during the cognitive task, despite no increase in perceived distraction or fatigue. Together, these findings suggest that repeated experiences of informational non-contingency can undermine the supervisory system that coordinates executive processes, leading to reduced cognitive regulation and engagement. By linking voice futility to cognitive demobilization, this study broadens the conceptualization of acquiescent silence from a motivational state to a cognitive risk factor, with implications for organizational functioning and psychological well-being.

1. Introduction

In hierarchical organizational contexts, individuals may come to perceive a lack of contingency between the act of speaking up and obtaining a meaningful change in their organizational environment (Pinder & Harlos, 2001). This perceived action-outcome non-contingency can lead individuals to withhold their input, not out of fear of speaking up but because of a sense of resignation (Pinder & Harlos, 2001; Van Dyne et al., 2003). This form of employee silence, referred to as acquiescent silence, reflects a passive withholding of voice based on the belief that speaking up is futile. Empirical studies and meta-analyses have consistently demonstrated that acquiescent silence is negatively associated with a range of individual attitudinal outcomes, including reduced organizational commitment, lower job satisfaction, and increased turnover intentions (e.g., Dedahanov & Rhee, 2015; Hao et al., 2022; Knoll et al., 2019; Zou et al., 2025). These relationships are relatively straightforward, as acquiescent silence conceptually aligns with psychological withdrawal and emotional disengagement (Pinder & Harlos, 2001), two phenomena that naturally erode individuals' affective ties to their work or organization.

In contrast, the relationship between acquiescent silence and task-related outcomes—such as performance decrements— has received comparatively less theoretical and empirical attention. Although existing meta-analytic evidence indicates that acquiescent silence is the form of silence most significantly associated with diminished task performance (Hao et al., 2022), the mechanisms underlying this relationship

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remain underexplored. Unlike attitudinal constructs, performance outcomes are shaped by complex interactions among cognitive, affective, and situational factors, rendering the explanatory pathway between acquiescent silence and performance less intuitively accessible.

To address this gap, the present research adopts a cognitive-motivational perspective to examine whether repeated experiences of voice non-instrumentality (i.e., the absence of voice impact despite efforts to propose inputs) affect individuals' cognitive functioning. Drawing on the information-processing reconceptualization of learned helplessness framework, we propose that persistent exposure to non-instrumental voice attempts fosters a sense of action-outcome non-contingency, which can reduce cognitive engagement and impair individuals' performance. In this study, we test whether such experiences lead to measurable impairment in individuals' executive functioning (comprising working memory, inhibitory control, and cognitive flexibility; Diamond, 2013). Executive functions constitute core cognitive processes that enable individuals to regulate their behavior and pursue goal-directed actions. By focusing on executive functioning, we aim to clarify why acquiescent silence may be associated with poorer task performance. While prior research has mainly linked performance decrements to antecedents of silence, such as abusive supervision or injustice perceptions (e.g., Lam & Xu, 2019; Whiteside & Barclay, 2013), we suggest that the experience of acquiescent silence itself may gradually undermine cognitive functioning. This perspective extends existing work by conceptualizing acquiescent silence not only as motivational withdrawal but also as a potential cognitive risk factor for task effectiveness.

1.1. Acquiescent silence as learned helplessness in organizations

The emergence of acquiescent silence in organizations can be usefully understood through the lens of the learned helplessness framework (Andrieu et al., 2024; Harlos & Knoll, 2021; Knoll et al., 2021; O'Donovan et al., 2021; Pinder & Harlos, 2001). This paradigm, originally developed within experimental psychology (Maier & Seligman, 1976, 2016; Peterson et al., 1993), posits that individuals who experience repeated failures to influence outcomes in their environment may come to expect that their actions are ineffective, leading them to disengage behaviorally and emotionally. Central to this process is the perception of non-contingency between one's actions and their consequences, a condition that has parallels in organizational life.

In many hierarchical work environments, employees may attempt to voice concerns or propose changes regarding their work context, only to find their input consistently disregarded by those in authority (Detert et al., 2013; Morrison, 2023; Morrison et al., 2015).When such experiences accumulate, they can foster a generalized belief that speaking up is futile, leading to a gradual withdrawal from voice engagement due to resignation rather than to fear or self-interest (Andrieu et al., 2024; Zou et al., 2025). From a sense of control perspective, these repeated experiences of non-contingency mirror the core conditions under which helplessness develops: A repeated mismatch between actions and outcomes that eventually results in passivity and disengagement. Crucially, this form of silence evolves progressively through unreciprocated interactions with authority figures or decision-making systems (Knoll, 2021; Schein, 2017).

While the emotional and attitudinal consequences of such disengagement (e.g., reduced organizational commitment or increased turnover intentions) are well-documented (Dedahanov & Rhee, 2015; Knoll et al., 2019), it is also important to consider how the experience of learned helplessness might affect broader employee cognitive functioning.

1.2. Learned helplessness and cognitive deficits

The informational reformulation of learned helplessness emphasizes its cognitive dimension (Bukowski & Kofta, 2017; Kofta, 1993; Kofta & Sędek, 1989, 1998; Sędek & Kofta, 1990). According to this view, prolonged exposure to uncontrollable events impairs subsequent task performance through cognitive exhaustion. Under controllable conditions, individuals test hypotheses, interpret feedback, and refine their strategies. Under uncontrollable conditions, however, feedback remains inconsistent or meaningless, preventing effective learning and gradually depleting cognitive resources. Over time, these futile efforts culminate in reduced cognitive mobilization and disengagement from effortful problem-solving. Empirical studies showed that induced informational learned helplessness disrupts subsequent performance in tasks involving either controlled attention, learning, planning, working memory, or cognitive flexibility (Bukowski et al., 2015, 2019; Gacek et al., 2017; von Hecker et al., 2000; von Hecker & Sędek, 1999).

This cognitive perspective aligns with models of executive functioning (Sparrow & Hunter, 2012), such as the Supervisory System model (Norman & Shallice, 1986). According to this framework, behavior is organized in hierarchically structured schemas that, with repetition, become automatized and triggered by environmental cues. Although such routines promote efficiency in familiar contexts, novel or ambiguous situations require a higher-order supervisory system to inhibit habitual responses and activate alternative strategies. This system underlies core executive functions (i.e., working memory, inhibitory control, and cognitive flexibility) which enable individuals to regulate behavior when environmental guidance is limited.

Within a learned helplessness framework, the supervisory system appears particularly vulnerable to deterioration under persistent action–outcome non-contingencies (Bukowski et al., 2024; Soral et al., 2021). When repeated cognitive adjustments fail to yield meaningful outcomes, the regulatory efficiency of this system declines as adaptive schemas receive insufficient reinforcement. Consequently, individuals may struggle to inhibit prepotent responses and to sustain engagement in novel, cognitively demanding tasks. Although executive functioning can be influenced by multiple factors (e.g., stress, affective states), learned helplessness provides a theoretically grounded mechanism through which repeated experiences of uncontrollability erode higher-order cognitive operations. From this standpoint, the supervisory-system perspective offers a coherent explanatory lens linking the motivational dynamics of uncontrollability with the executive deficits consistently observed in helplessness research.

Extending this reasoning to organizational contexts, when repeated voice attempts are met with rejections, conditions of informational non-contingency arise. Despite sustained cognitive efforts to speak up (e.g., planning, framing, assessment of risks, strategic monitoring; Detert & Treviño, 2010), employees may fail to perceive a reliable connection between speaking up and achieving meaningful change. When these resource-intensive efforts repeatedly yield no effect, the sense of control that sustains proactive cognition may erode. As a result, employees may not only withdraw behaviorally through acquiescent silence but also experience a broader depletion of executive functioning, manifested in reduced working memory, weakened inhibitory control, and diminished cognitive flexibility.

1.3. Study overview

Building on this theoretical framework, which posits that repeated experiences of uncontrollability impair the supervisory processes underlying executive functioning, the present study examines whether repeated exposure to non-instrumental voice produces measurable decrements in executive performance, analogous to those observed in learned helplessness research.

Consistent with major models of executive functioning, we focused on the three cognitive components most relevant to adaptive, goal-directed behavior: working memory, inhibitory control, and cognitive flexibility (Banich, 2009; Diamond, 2013; Sambol et al., 2023; Zelazo, 2015). Working memory refers to the ability to maintain and manipulate task-relevant information; inhibitory control denotes the capacity to

suppress automatic or irrelevant responses; and cognitive flexibility reflects the ability to shift between rules, perspectives, or task sets. Together, these top-down processes enable individuals to regulate thought and behavior in novel or demanding contexts (Zelazo et al., 2004).

Executive performance was assessed in our study using the Parametric Go/No-Go (PGNG) task, an instrument used in clinical and experimental research to evaluate executive functioning and processing speed (Langenecker et al., 2007; Langenecker & Nielson, 2003; Votruba & Langenecker, 2013). The PGNG provides an integrated assessment of sustained attention, inhibition, shifting, updating, and processing speed (see Materials section for details).

Accordingly, this study tests whether repeated experiences of voice non-instrumentality lead to measurable impairments in executive functioning. By examining the cognitive costs of sustained informational non-contingency, we aim to provide a more comprehensive understanding of how organizational silence undermines performance, not only through reduced motivation or engagement, but through the depletion of the executive resources required for effective task execution.

2. Method

2.1. Transparency and openness

The study was pre-registered[1] and data are available in an OSF depository.[2] Analytic codes, and additional materials for the experiment are available upon request.

2.2. Participants

Participants were recruited from a university in southern France through mailing lists and poster advertisements. The study was conducted in a controlled laboratory environment to ensure standardized testing conditions.

An a priori power analysis conducted with G*Power (version 3.1.9.7.) indicated that a minimum of $N = 159$ participants would be required to detect a medium-sized between-group effect ($f = 0.25$, $\alpha = 0.05$, power = 0.80) for the three experimental conditions (Low Voice Impact, High Voice Impact, Control). This analysis did not take into account the within-subject factor corresponding to the three consecutive levels of the PGNG task, as our aim at the preregistration stage was to adopt a conservative estimation based on the primary between-group manipulation. A total of 131 participants were initially recruited. Two participants were excluded from the analysis due to failure to provide suggestions during the initial phase of the experimental manipulation (details provided below). The final sample consisted of 129 participants, with an average age of 21.5 years ($SD = 4.14$). The majority of participants were female (85.3 %).

Although the achieved sample size was slightly below the a priori target, a post hoc sensitivity check indicated that, given the mixed 3 (Group: Low Voice Impact, High Voice Impact, Control) × 3 (PGNG Level: 1, 2, 3) design, the study retained approximately 86 % power to detect medium-sized Group × Level interaction effects (assuming $\rho = 0.5$). This level of sensitivity is consistent with or exceeds that of prior experimental studies examining the cognitive consequences of uncontrollability using comparable paradigms (e.g., Bukowski et al., 2015, 2019), many of which reported reliable results with sample sizes between 75 and 100. Thus, the present sample provides sufficient statistical power for detecting theoretically meaningful effects while remaining consistent with established research practices in this domain.

2.3. Material

2.3.1. Voice impact manipulation: acquiescent silence learning task (ASL-T)

The ASL-T was designed to manipulate participants' perceived impact on relevant voice-related decisions, inspired by the learned helplessness protocol (Andrieu et al., 2024). In this task, participants were asked to generate suggestions on four distinct topics related to student life at their university (e.g., "ideas for welcoming new students"). They were informed that their input could be added to a national consultation aimed at gathering student feedback for potential improvements. For each of the four topics, participants could submit up to 10 suggestions, with the instructions to provide unique ideas in each space. Participants were told that once they voiced their opinions on a topic, their input would be immediately examined by a reviewer, purportedly a member of the research team, and feedback would be provided regarding whether their suggestions were retained for the study. In reality, the feedback messages were pre-recorded and scripted, tailored to the participant's assigned experimental condition. In the low voice impact condition (LVI; $n = 43$), suggestions were systematically rejected, creating a repeated occurrence of action-outcome non-contingencies (i.e., voice non-instrumentality). In the high voice impact condition (HVI; $n = 42$), suggestions were systematically accepted, leading participants to believe their input was instrumental and impactful. In the control condition ($n = 44$), no feedback was provided regarding the fate of the suggestions; participants were simply instructed to move on to the next topic without knowing whether their suggestions are retained. After completing their propositions on a topic, participants viewed a "loading" screen to simulate a delay while the experimenter supposedly reviewed the responses (from 18 s to 36 s). Then the feedback message appropriate for the experimental condition was presented.

2.3.2. Cognitive task: parametric go/no-go (PGNG)

The Parametric Go/No-Go (PGNG; Langenecker et al., 2007) was implemented using PsychoPy (version 2021.2.3). This task involved the fast and sequential presentation of individual lowercase letters (white font on a gray background, Open Sans typeface, displayed at one-quarter of the screen height), each shown for 600 ms with no inter-stimulus interval. Participants were instructed to press a response button when they saw specific target letters —X, Y, or Z— and avoid pressing the button for all the other non-target letters depending on the rule set for each of three levels (Level 1 to Level 3; see Fig. 1; Appendix A). The three levels of the task were designed to elicit varying patterns of executive involvement rather than to represent linear increases in task difficulty. As described by Langenecker et al. (2007), each level engages distinct but overlapping executive processes and affords different opportunities for strategic regulation.

In Level 1, participants were told to press the response button every time one of the target letters appeared on the screen. This condition, which can be considered a baseline "Go" level, primarily engages vigilance, sustained attention, and processing speed. Targets and distractors remained constant throughout this level. Two performance metrics were recorded: Percentage of Correct Responses to targets (PCT) and Efficiency, calculated as a ratio of accuracy to reaction time.[3]

In Level 2, the task introduced a non-repetition rule, and the letter Z was removed from the stimulus set. Participants now had to keep track of the last target letter (either X or Y) they responded to, as that letter became a lure until the other target letter appeared. In other words, once participants responded to an X, they were required to withhold their

[1] https://doi.org/10.17605/OSF.IO/VGMCW.
[2] https://doi.org/10.17605/OSF.IO/W5GQ8.

[3] Efficiency = (PCT/mean response time) * 100.

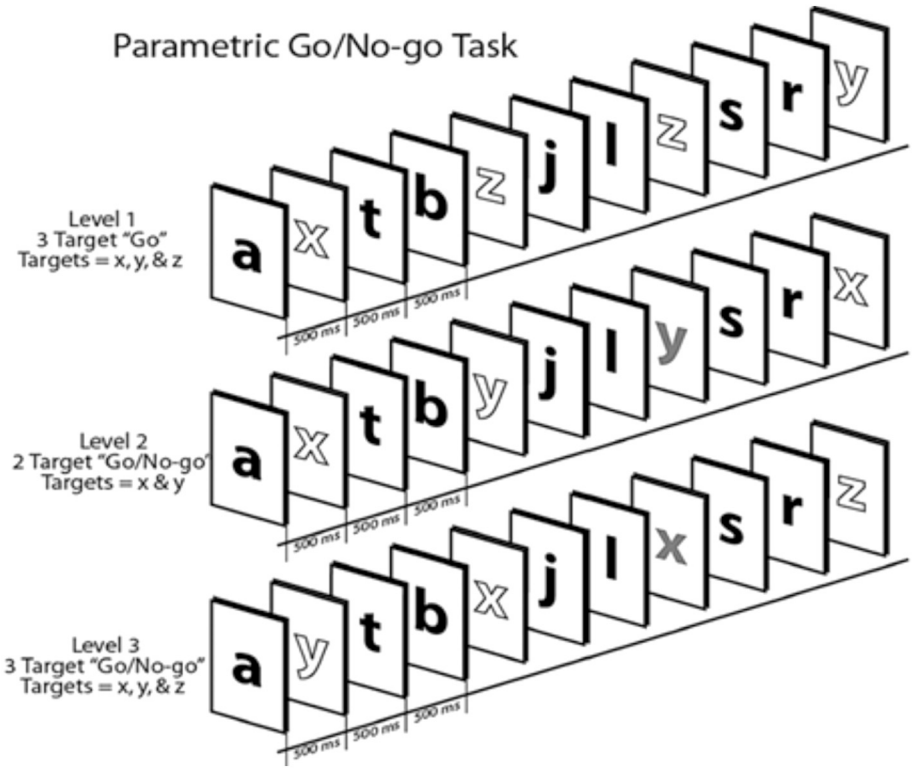

Fig. 1. Illustration of the three levels of the PGNG.
Note. From Votruba and Langenecker (2013). Hollowed letters represent a correct "Go" responses (when the participant must press the button for one of the target letters). Gray letters represent lures, which are correct "No-Go" responses (when the participant must not press the button and inhibit the predominant response for one repeated target letter). Black and bold letters represent all flashed letters except the three target letters and the lures.

response if X reappeared before a Y had been presented, and vice versa. This level engages sustained attention, context-based inhibition, and cognitive flexibility (i.e., set-shifting). In addition to PCT and Efficiency,[4] a third performance measure was collected: Percentage of Correct Inhibition (PCI) to lures, defined as the proportion of correct non-responses when a repeated target appeared in violation of the non-repetition rule. In this second level, participants may have greater success compared to levels 1 and 3 due to the implementation of a strategy: tracking only two working-memory targets, making anticipation and recall easier than in levels 1 or 3 (Langenecker et al., 2007). In this manner, switching between two targets at level 2 is likely less demanding than maintaining three in working memory, possibly leading to better accuracy and response time in level 2 than in level 1.

In Level 3, participants were instructed to follow the same non-repetition rule as in Level 2, but with the reintroduction of Z as a third target. This increases the task's complexity, as the non-repetition rule now applies across all three target letters (X, Y, and Z). After responding to any one of the targets, participants were required to inhibit a response to repeated presentations of that same target unless both other targets had subsequently appeared. As a result, this level places greater demands on contextual updating, response inhibition, and working memory capacity. The addition of the third target reduces the participants' ability to use simple response strategies and requires continuous tracking of the target sequence.

Metrics measured during the three levels of the task reflect participants' executive performance across different domains: sustained attention, shifting and updating (PCT), inhibitory control (PCI), and Efficiency, a proxy for global executive functioning (Langenecker et al., 2007; Votruba & Langenecker, 2013).

At each level, participants were instructed to respond as quickly and accurately as possible according to the task rules. Prior to each level, written instructions and a brief practice session were provided to ensure comprehension. All participants completed the three levels in the same fixed order.

2.4. Procedure and measures

Upon arrival, participants were seated in individual testing booths and instructed to read and sign the informed consent form. The experimental session consisted of two main phases, presented as two distinct and unrelated studies to minimize demand effects.

In the first phase, participants completed the ASL-T after being randomly assigned to an ASL-T condition. No manipulation check was administered during this task for three main reasons. First, direct checks —especially self-report items— can act as interventions in themselves, potentially weakening or even reversing the intended effects of the manipulation (Ejelöv & Luke, 2020; Hauser et al., 2018). Second, introducing a manipulation check could have compromised the cover story that the two phases of the session were unrelated, thereby increasing the risk of demand characteristics. Third, prior work using the exact same paradigm with a comparable participant population (Andrieu et al., 2024) has already demonstrated that the low voice impact (LVI) condition reliably leads to lower perceived voice impact and increased self-reported feelings of helplessness.

In the second phase, participants completed the PGNG task. This task allowed us to assess executive functioning across three different levels. After the completion of each PGNG level, participants rated their subjective experience by responding to six self-report items: Three items, adapted from Kiss and Linnell (2021), respectively assessed perceived

[4] In Level 2 and Level 3, the accuracy scores used to compute efficiency correspond to a ratio between PCT and PCI adjusted for the proportions of targets and lures: ((Ratio of targets * PCT: ratio of lures * PCI) / sum of ratios) * 100.

disturbance during the task, perceived concentration while doing the task, and self-reported exhaustion. The three remaining items evaluated perceived control over performance, motivation to persist, and perceived competence. Responses were recorded on a 7-point Likert scale ranging from −3 ("strongly disagree") to +3 ("strongly agree"). All items are provided in Appendix B.

Upon completion of the experimental session, participants were debriefed and provided with a full explanation of the study's objectives.

3. Results

All analyses were conducted using R statistical software (version 4.2.2; R Core Team, 2021). All dependent variables were analyzed using general linear models (GLMs). Each model included two factors: 3, ASL-T condition (LVI, HVI, or control) * 3, PGNG level (Levels 1, 2, and 3), along with their interaction. Because differential effects across task difficulty levels were not preregistered, all analyses involving the Level factor should be considered exploratory rather than confirmatory. Post hoc comparisons were performed using the *emmeans* package (Lenth, 2021) with Holm correction.

3.1. Perceived distraction while doing the task

We first analyzed perceived distraction to ensure that participants were not significantly disturbed while completing the cognitive task. Results revealed no significant main effect of voice impact condition, $F(2, 378) = 0.11$, $p = 0.894$, no effect of PGNG level, $F(2, 378) = 0.41$, $p = 0.662$, and no significant interaction between the two, $F(4, 378) = 0.36$, $p = 0.831$. These findings suggest that participants did not perceive themselves as being distracted by internal or external factors during the task.

3.2. Task performance (PGNG)

As several core dependent variables deviated from normality (see Table 1), we applied a non-paranormal transformation using the *huge.npn* function from the *huge* package (Zhao et al., 2012) to normalize the data.[5] Results are graphically summarized in Fig. 2.

3.2.1. Percentage of correct responses (PCT)

A significant main effect emerged for both the voice impact condition, $F(2, 378) = 4.92$, $p = 0.008$, $\eta^2 = 0.021$, and the PGNG level, $F(2, 378) = 39.39$, $p < 0.001$, $\eta^2 = 0.164$, as well as for their interaction, $F(4, 378) = 3.36$, $p = 0.010$, $\eta^2 = 0.028$. Post hoc comparisons showed the group difference to be significant during level 2 of the PGNG only: Participants in the LVI condition performed worse ($M = 66.7$, $SD = 24.7$) than those in the HVI condition ($M = 83.6$, $SD = 13.0$), $t = 4.75$, $p_{holm} < 0.001$, and in the control group ($M = 75.4$, $SD = 19.5$, $t = 2.39$, $p_{holm} = 0.033$). No group differences were observed in Levels 1 or 3 (all $ps > 0.999$).

3.2.2. Percentage of correct inhibitions (PCI)

The analyses showed main effects of the voice impact condition, $F(2, 252) = 26.15$, $p < 0.001$, $\eta^2 = 0.140$, and the PGNG level, $F(1, 252) = 66.52$, $p < 0.001$, $\eta^2 = 0.177$, but no significant interaction, $F(2, 252) = 1.85$, $p = 0.159$. Comparisons showed that participants in the LVI condition inhibited significantly fewer responses ($M = 63.8$, $SD = 20.9$) than those in the HVI ($M = 81.1$, $SD = 12.8$), $t = 6.76$, $p_{holm} < 0.001$, and control conditions ($M = 75.4$, $SD = 13.8$), $t = 5.61$, $p_{holm} < 0.001$. No difference was found between HVI and control groups ($t = 1.22$, $p_{holm} = 0.223$). Globally, across all groups, PCI declined significantly from Level 2 to Level 3, $t = 8.22$, $p < 0.001$, confirming the increased level of inhibition in the final PGNG level.

3.2.3. Efficiency

Efficiency analyses mirrored the pattern observed for PCT: The analyses revealed significant main effects of voice impact condition, $F(2, 378) = 5.99$, $p = 0.002$, $\eta^2 = 0.024$, and PGNG level, $F(2, 378) = 47.13$, $p < 0.001$, $\eta^2 = 0.19$, as well as a significant interaction between both, $F(4, 378) = 2.84$, $p = 0.024$, $\eta^2 = 0.022$. Post hoc comparisons revealed the group difference to be significant in Level 2 only: Participants in the LVI group were less efficient ($M = 15.8$, $SD = 5.07$) than those in the HVI group ($M = 19.5$, $SD = 3.16$), $t = 4.69$, $p_{holm} < 0.001$, and in the control condition ($M = 17.4$, $SD = 4.32$, $t = 2.73$, $p_{holm} = 0.013$. No group differences were observed for Levels 1 or 3 (all $ps > 0.999$).

3.2.4. Measuring scales

Analyses of self-reported exhaustion and perceived concentration revealed significant main effects of PGNG level only: $F(2, 378) = 4.92$, $p = 0.007$, $\eta^2 = 0.025$ for exhaustion, and $F(2, 378) = 8.36$, $p < 0.001$, $\eta^2 = 0.042$ for concentration. Post hoc comparisons indicated that participants felt more exhausted after Level 3 of the PGNG ($M = 0.54$, $SD = 1.85$) compared with Level 1 ($M = -0.12$, $SD = 1.76$; $t = -3.11$, $p_{holm} = 0.006$). In contrast, participants reported lower concentration after Level 3 ($M = 1.44$, $SD = 1.73$) than after Level 1 ($M = 2.03$, $SD = 1.15$; $t = 4.07$, $p_{holm} < 0.001$).

Analyses of the three remaining subjective measures revealed a significant main effect of the voice impact manipulation on reported motivation in the task, $F(2, 378) = 13.02$, $p < 0.001$, $\eta^2 = 0.062$, with participants in the LVI condition reporting significantly lower motivation ($M = -0.30$, $SD = 0.93$) than those in the HVI ($M = 0.20$, $SD = 0.75$, $t = 5.02$, $p_{holm} < 0.001$) and control conditions ($M = -0.006$, $SD = 0.82$, $t = 2.28$, $p_{holm} = 0.023$). Reported motivation was also significantly affected by the PGNG levels, $F(2, 378) = 7.48$, $p < 0.001$, $\eta^2 = 0.031$, with significantly lower reported motivation in Level 3 ($M = 0.94$, $SD = 1.83$) compared to Level 1 ($M = 1.78$, $SD = 1.48$; $t = 3.86$, $p_{holm} < 0.001$) over all experimental conditions.

A significant main effect of both the voice impact manipulation, $F(2, 378) = 4.53$, $p = 0.011$, $\eta^2 = 0.022$ and the PGNG levels, $F(2, 378) = 9.98$, $p < 0.001$ was also found on perceived control over the task. Post hoc comparisons indicated lower ratings of perceived control for participants in the LVI condition ($M = -0.14$, $SD = 0.88$) when compared to participants in the HVI condition ($M = 0.19$, $SD = 0.94$, $t = 2.95$, $p_{holm} = 0.10$). Regarding PGNG levels, perceived control decreased significantly across levels: participants reported higher perceived control after Level 1 ($M = 0.85$, $SD = 1.53$) compared with Level 2 ($M = 0.21$, $SD = 1.81$; $t = 2.00$, $p_{holm} = 0.046$) and Level 3 ($M = -0.35$, $SD = 1.81$; $t = 4.46$, $p_{holm} < 0.001$). Perceived control was also higher after Level 2 compared with Level 3 ($t = 2.46$, $p = 0.029$).

Finally, only a main effect of PGNG level was observed on reported perceived competence while doing the task, $F(2, 378) = 17.98$, $p < 0.001$, with all participants feeling less competent at Level 3 ($M = -0.56$, $SD = 1.85$) compared to Level 1 ($M = 1.00$, $SD = 1.57$, $t = 5.99$, $p < 0.001$) and Level 2 ($M = 0.22$, $SD = 1.80$, $t = 2.79$, $p = 0.005$). Participants also reported feeling less competent at Level 2 compared to Level 1, $t = 3.20$, $p = 0.005$.

4. Discussion

The present study provides initial experimental evidence that repeated exposure to rejected voice attempts—operationalized here as systematically unendorsed suggestions in a simulated consultation task— can impair executive functioning on a subsequent demanding cognitive task. Drawing on the informational learned helplessness model, we proposed that repeated experiences of informational non-contingency would lead to reduced efficiency in executive processes. These findings extend theoretical perspectives on acquiescent silence by

[5] This semiparametric transformation uses nonparanormal copulas to model multivariate Gaussian dependence structures while preserving nonparametric marginal distributions (Liu et al., 2012, 2009).

Table 1
Raw means of the dependent variable by voice impact manipulation condition and PGNG Level.

	Dependent variable								
	PCT			PCI			Efficiency		
Levels	1	2	3	1	2	3	1	2	3
Conditions									
LVI	75.5[a]	66.7[a]	59.4	–	69.3[a]	58.3	17.0[a]	15.8[a]	13.1
HVI	78.2	83.6[a]	60.0	–	89.3[a]	72.9[a]	17.4	19.5	13.7
Control	78.4[a]	75.4	61.7[a]	–	85.4[a]	71.2	17.3[a]	18.0[a]	13.8

[a] W = Shapiro-Wilk test significant at $p < 0.05$ for the raw data distributions.

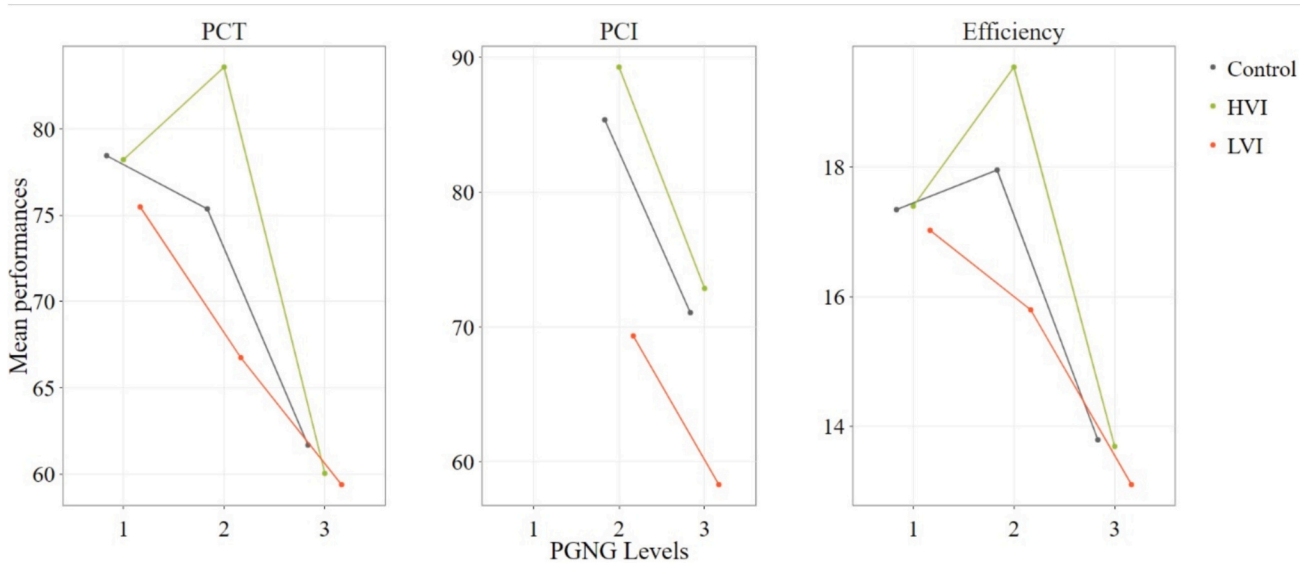

Fig. 2. Graphical representation of PGNG measures depending on the voice impact manipulation and task level.
Note. "Control" = Control group; "HVI" = High Voice Impact, "LVI" = Low Voice Impact. Numbers 1, 2 and 3 on the x axis represent levels of the PGNG task.

suggesting that its consequences may go beyond attitudinal or motivational domains to affect core cognitive processes involved in adaptive task performance.

4.1. Theoretical implications

These results advance the broader literature on silence and cognitive functioning in several ways. First, they provide initial experimental support for the conceptual parallel between acquiescent silence and learned helplessness, a link that has been theorized but rarely tested empirically (Andrieu et al., 2024; O'Donovan et al., 2021; Pinder & Harlos, 2001; Sherf et al., 2021). The observed performance decrements in the low voice impact condition suggest that repeated non-instrumental voiced input may undermine executive functioning in contexts requiring sustained cognitive effort. However, as this study used a student sample in a controlled laboratory setting, caution is warranted before generalizing directly to organizational environments.

These findings contribute to the idea that performance deficits related to silence are not solely byproducts of antecedent conditions (e.g., abusive supervision) but may also stem from the emergent cognitive and motivational consequences of unacknowledged voice attempts themselves. While further research in organizational field settings is needed, these results point to the possibility that repeated voice futility experiences may degrade individuals' cognitive performance on tasks over time.

Exploratory analyses revealed that group differences emerged specifically at Level 2 of the PGNG. Rather than reflecting simple task difficulty, this pattern may be linked to the nature of the executive control required at this level. As described in the Methods section, Level 2 introduces a non-repetition rule involving two target letters, which allows participants to adopt a strategy based on tracking and anticipation. In this manner, participants can easily maintain only one letter in working memory (X or Y) and alternatively change the target by following the non-repetitive rule. If this strategy limits cognitive costs by decreasing the involvement of inhibition and the working memory saturation, the participants must remain vigilant and reactive to the visual stimuli. For participants in the low voice impact (LVI) condition, however, repeated exposure to voice non-instrumentality may have depleted these regulatory resources, making it more difficult to maintain or deploy an adaptive strategy. From the perspective of the Supervisory Attentional System model (Norman & Shallice, 1986), these effects may reflect a temporary weakening of higher-order executive control processes responsible for inhibiting dominant responses, updating task rules, and flexibly adjusting behavior in non-routine contexts. This interpretation is consistent with informational learned helplessness theory (Bukowski & Kofta, 2017; Kofta & Sędek, 1998; Sędek & Kofta, 1990), which predicts that under sustained non-contingency, individuals have impaired ability to regulate cognition in response to situational demands. The absence of group differences at Level 3 likely reflects a convergence toward uniformly low performance, suggesting that the task's difficulty may have masked potential group effects: At this stage, the reintroduction of a third target letter dramatically increases contextual updating and working-memory load, reducing the possibility of effective strategic control for all participants. As such, Level 2 may represent the condition most sensitive to detecting impairments in adaptive executive regulations, that is, when the task demands are high enough to require strategic engagement but not so high as to induce general overload.

Taken together, the present findings provide initial support for our theoretical rationale, as performance decrements emerged precisely under task conditions that required sustained executive regulation and strategic control. However, the use of the PGNG task necessarily narrows the operationalization of executive functioning to processes primarily related to inhibitory control, updating, and attentional switching under varying cognitive load. Although these components are central to the Supervisory Attentional System, they do not exhaust the range of cognitive processes potentially affected by learned helplessness. Future studies could therefore extend this work by incorporating complementary cognitive paradigms (e.g., tasks assessing planning, rule switching, complex problem solving) to examine whether the effects of voice non-instrumentality generalize across a broader spectrum of executive domains.

Another revealing pattern emerged when comparing the low voice impact (LVI) group to the control condition in which participants received no feedback about the fate of their voiced input. Despite the absence of acknowledgment in both conditions, only participants in the LVI group —who experienced consistent rejection— exhibited significant impairments in executive functioning. This divergence may suggest that the cognitive costs associated with voice futility are not simply a matter of being ignored or left in the dark, but stem from repeated, explicit disconfirmation.

Whereas the main analyses focused on objective indicators of executive functioning, the results for the subjective measures provide complementary insights. Participants in the low voice impact (LVI) condition reported significantly lower motivation to persist and reduced perceived control over the cognitive task. These self-reports parallel the performance deficits observed on Level 2 of the PGNG task indicating that participants' experience of diminished control and effort aligned with measurable cognitive impairments. This pattern is consistent with the dual-component view of informational learned helplessness (Bukowski & Kofta, 2017; Kofta & Sędek, 1998; Sędek & Kofta, 1990), which posits that uncontrollability impairs both executive regulation (cognitive demobilization) and the willingness to invest effort (motivational withdrawal). From this perspective, the deficits observed here may reflect a combined effect: Participants' reduced capacity to sustain adaptive cognitive strategies was likely compounded by a decrease in perceived control and goal-directed effort. However, because we did not directly measure invested effort or emotional reactions, we cannot determine whether these performance decrements resulted from reduced motivation, from cognitive inefficiency despite effort (Ric & Scharnitzky, 2003), or from their interaction. Importantly, perceived competence did not differ significantly across conditions, suggesting that participants did not feel less capable overall, but rather experienced a more specific reduction in control and effort mobilization. Future studies should therefore incorporate validated measures of perceived mental effort and cognitive exhaustion (e.g., Sędek & Kofta, 1990) to disentangle more precisely the cognitive and motivational pathways underlying these effects.

4.2. Limitations and future directions

Despite being able to assess causal relations through experimental design and theory-based hypotheses, several limitations of our study should be noted. Most importantly, the experimental procedure, conducted in a controlled laboratory setting with university students, lacks ecological validity relative to real-world organizational contexts. The task and voice feedback manipulations, while allowing precise control and measurement, simplify the complex social dynamics and emotional factors present in actual workplaces. This limits the extent to which findings can be directly generalized to organizational settings. Future research should seek to replicate and extend these results in more naturalistic environments and with occupationally diverse samples to better capture the nuances of employee voice experiences.

Regarding the experimental protocol, a notable limitation is the absence of a direct manipulation check. This decision was deliberate: Direct self-report measures could have interfered with our manipulations and compromised the cover story linking the task phases (Ejelöv & Luke, 2020; Hauser et al., 2018). However, this choice also limits our ability to disentangle potential affective or motivational processes (e.g., self-esteem threat, perceived exclusion) that may co-occur with informational helplessness. Future research could complement this approach by combining behavioral or physiological indicators of cognitive control with post-hoc assessments of perceived control or affective states, to better disentangle the cognitive and emotional dynamics underlying unacknowledged voice.

In addition, the sample consisted predominantly of young adult students, the majority of whom were female, which restricts the demographic diversity of the study. This homogeneity may limit the generalizability of the findings to broader working populations that vary by age, gender, socio-economic status, and cultural background. Future studies should strive to include more demographically and professionally diverse participants to examine whether the observed cognitive and motivational effects of voice rejection hold across different occupational contexts and population subgroups.

Although the Parametric Go/No-Go task provides a well-established measure of executive functioning, it primarily assesses inhibitory control, updating, and attentional switching under varying levels of cognitive load. Incorporating complementary paradigms targeting other cognitive—such as planning, complex problem-solving, or divided attention—alongside physiological indices (e.g., EEG, pupillometry) could further clarify the specific cognitive and affective mechanisms underlying the effects of voice non-instrumentality. In addition, because the moderation of the experimental effect by task difficulty level was not preregistered, this pattern should be interpreted cautiously and considered exploratory, pending direct preregistered replication. Future preregistered replication studies will be essential to confirm the robustness and generalizability of these difficulty-dependent effects.

Finally, future work could examine the potential reversibility of cognitive demobilization linked to repeated voice rejection. Investigating whether interventions that enhance perceived control or voice efficacy restore executive functioning would be valuable. Longitudinal designs would also be important to trace how episodic experiences of voice non-endorsement evolve into chronic acquiescent silence and their cumulative cognitive impact over time. Future preregistered replication studies will be essential to confirm the robustness and generalizability of these difficulty-dependent effects.

4.3. Practical implications

From a practical standpoint, these findings carry warnings for organizational leaders and policymakers. Repeatedly rejected employee input —even without hostility— can quietly erode not only morale and motivation but also cognitive functioning. As executive functions are essential for planning, prioritization, learning, and interpersonal coordination, their erosion can have cascading effects on team performance, innovation, and workplace safety.

Executive functions, such as planning, attention, and cognitive flexibility, are critical in many everyday roles — whether in student performance, civic engagement, or professional tasks. Their deterioration can have cascading effects, such as reduced learning, innovation, coordination, or responsiveness. These insights highlight the need to reassess how feedback and participatory mechanisms are managed. Interventions aimed at improving transparency, responsiveness to contributions, and closing the feedback loop may serve a dual purpose: sustaining engagement and protecting cognitive performance. Such strategies are particularly vital in high-stakes or cognitively demanding environments — such as education, healthcare, public administration, and emergency services — where executive functioning underpins both individual and collective effectiveness.

5. Conclusion

This study sheds light on a previously underexplored cognitive mechanism through which non-instrumental voice attempts may impair individual functioning across various participatory contexts. By experimentally demonstrating that repeated informational non-contingency leads to diminished executive functioning, we offer an extension to the silence literature: silence, especially of the acquiescent kind, is not only a social or motivational issue but also a cognitive one.

To foster healthier and more effective systems — whether in organizations, educational institutions, public services, or other authority-driven contexts — it is imperative to recognize the cognitive cost of neglecting the voices of those who seek to contribute constructively.

CRediT authorship contribution statement

Clement F.A. Andrieu: Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization, Writing – review & editing, Writing – original draft. **Roxane L. Bartoletti:** Methodology, Conceptualization, Writing – original draft. **Isabelle Milhabet:** Supervision, Conceptualization, Writing – original draft. **Ambre Denis-Noël:** Visualization, Software, Formal analysis, Writing – original draft. **Dirk D. Steiner:** Supervision, Conceptualization, Writing – original draft.

Ethics statement

The study was approved by the ethical committee of the authors' university (decision number 2023–02). Informed consent was obtained from all individual participants included in the study.

Declaration of competing interest

The authors have no conflicts of interest to disclose. This work did not benefit from any financial support.

Appendix A

A.1. PGNG levels descriptions

A.1.1. Level 1

Participants were instructed to respond to the letters *x*, *y*, and *z*, regardless of their order or repetition. This level consisted of 33 target letters and 192 non-target letters, randomly ordered for each participant.

A.1.2. Level 2

Participants were again asked to identify *x* and *y* as targets, but only if the current target letter differed from the last one they responded to. That is, they had to alternate between *x* and *y* (e.g., respond to *y* only if the previous target was *x*, and vice versa), requiring them to inhibit responses to consecutive identical targets. This level comprised 50 valid target letters, 16 lure trials (repeated targets to be inhibited), and 264 non-target letters.

A.1.3. Level 3

The final level added *z* as a third alternating target, requiring participants to apply the same alternation rule among three letters (*x*, *y*, *z*). As with Level 2, participants needed to monitor and inhibit responses to repeated target letters. This level included 50 valid target letters, 15 lure trials, and 264 non-target letters.

Appendix B

B.1. Item list

Self-reported exhaustion:

- "During the task, I felt tired, my mind was blank, or my thoughts were elsewhere."

Perceived concentration while doing the task:

- "I was focused on the task or on how I was performing it."

Perceived disturbance during the task:

- "During the task, I was thinking about things around me in the room (objects, people, sounds, temperature, etc.) or about sensations I was feeling in my body (hunger, thirst, pain)."

Motivation to persist:

- "I stayed motivated to do the task from beginning to end."

Perceived control over performance:

- "I felt in control during the task from beginning to end."

Perceived competence:

- "I felt competent in performing the task."

Data availability

The data that support the findings of this study are available on OSF at https://doi.org/10.17605/OSF.IO/W5GQ8.

References

Andrieu, C. F. A., Milhabet, I., Denis-Noël, A., & Steiner, D. D. (2024). Voice in the void: From voice to acquiescent silence over time as learned helplessness in organizations. *Journal of Work and Organizational Psychology, 40*(2), 103–118. https://doi.org/10.5093/jwop2024a9

Banich, M. T. (2009). Executive function: The search for an integrated account. *Current Directions in Psychological Science, 18*(2), 89–94. https://doi.org/10.1111/j.1467-8721.2009.01615.x

Bukowski, M., Asanowicz, D., Marzecová, A., & Lupiáñez, J. (2015). Limits of control: The effects of uncontrollability experiences on the efficiency of attentional control. *Acta Psychologica, 154*, 43–53. https://doi.org/10.1016/j.actpsy.2014.11.005

Bukowski, M., de Lemus, S., Marzecová, A., Lupiáñez, J., & Gocłowska, M. A. (2019). Different faces of (un)controllability: Control restoration modulates the efficiency of task switching. *Motivation and Emotion, 43*(1), 12–34. https://doi.org/10.1007/s11031-018-9745-8

Bukowski, M., & Kofta, M. (2017). From coping to helplessness: Effects of control deprivation on cognitive and affective processes. In M. Bukowski, I. Fritsche, A. Guinote, & M. Kofta (Eds.), *Coping with lack of control in a social world* (pp. 15–34). Routledge.

Bukowski, M., Śpiewak, S., Różańska, A., & Gruszka, A. (2024). Personal control and cognitive flexibility: Does uncontrollability reduce flexible shifting of attention? *Learning and Motivation, 87*, Article 102024. https://doi.org/10.1016/j.lmot.2024.102024

Dedahanov, A. T., & Rhee, J. (2015). Examining the relationships among trust, silence and organizational commitment. *Management Decision, 53*(8), 1843–1857. https://doi.org/10.1108/MD-02-2015-0041

Detert, J. R., Burris, E. R., Harrison, D. A., & Martin, S. R. (2013). Voice flows to and around leaders: Understanding when units are helped or hurt by employee voice. *Administrative Science Quarterly, 58*(4), 624–668. https://doi.org/10.1177/0001839213510151

Detert, J. R., & Treviño, L. K. (2010). Speaking up to higher-ups: How supervisors and skip-level leaders influence employee voice. *Organization Science, 21*(1), 249–270. https://doi.org/10.1287/orsc.1080.0405

Diamond, A. (2013). Executive functions. *Annual Review of Psychology, 64*(1), 135–168. https://doi.org/10.1146/annurev-psych-113011-143750

Ejelöv, E., & Luke, T. J. (2020). "Rarely safe to assume": Evaluating the use and interpretation of manipulation checks in experimental social psychology. *Journal of Experimental Social Psychology, 87*, Article 103937. https://doi.org/10.1016/j.jesp.2019.103937

Gacek, M., Smoleń, T., & Pilecka, W. (2017). Consequences of learned helplessness and recognition of the state of cognitive exhaustion in persons with mild intellectual disability. *Advances in Cognitive Psychology, 13*(1), 42–51. https://doi.org/10.5709/acp-0205-6

Hao, L., Zhu, H., He, Y., Duan, J., Zhao, T., & Meng, H. (2022). When is silence golden? A meta-analysis on antecedents and outcomes of employee silence. *Journal of Business and Psychology, 37*(5), 1039–1063. https://doi.org/10.1007/s10869-021-09788-7

Harlos, K. P., & Knoll, M. (2021). Employee silence and workplace bullying. In P. D'Cruz, E. Noronha, E. Baillien, B. Catley, K. P. Harlos, A. Hogh, & E. G. Mikkelsen (Eds.), *2. Pathways of job-related negative behavior* (pp. 202–229). Springer.

Hauser, D. J., Ellsworth, P. C., & Gonzalez, R. (2018). Are manipulation checks necessary? *Frontiers in Psychology, 9*. https://doi.org/10.3389/fpsyg.2018.00998

Kiss, L., & Linnell, K. J. (2021). The effect of preferred background music on task-focus in sustained attention. *Psychological Research, 85*(6), 2313–2325. https://doi.org/10.1007/s00426-020-01400-6

Knoll, M. (2021). Silence in organizations – What we need to know, what we know, and what we don't know. In S. Starystach, & K. Höly (Eds.), *Silence of organizations: How organizations cover up wrongdoings* (pp. 17–42). Heidelberg.

Knoll, M., Hall, R. J., & Weigelt, O. (2019). A longitudinal study of the relationships between four differentially motivated forms of employee silence and burnout. *Journal of Occupational Health Psychology, 24*(5), 572–589. https://doi.org/10.1037/ocp0000143

Knoll, M., Neves, P., Schyns, B., & Meyer, B. (2021). A multi-level approach to direct and indirect relationships between organizational voice climate, team manager openness, implicit voice theories, and silence. *Applied Psychology. An International Review, 70*(2), 606–642. https://doi.org/10.1111/apps.12242

Kofta, M. (1993). Uncertainty, mental models, and learned helplessness: An anatomy of control loss. In G. Weary, F. Gleicher, & K. L. Marsh (Eds.), *Control motivation and social cognition* (pp. 122–153). Springer. https://doi.org/10.1007/978-1-4613-8309-3_5.

Kofta, M., & Sędek, G. (1989). Repeated failure: A source of helplessness or a factor irrelevant to its emergence? *Journal of Experimental Psychology, 118*, 3–12. https://doi.org/10.1037//0096-3445.118.1.3

Kofta, M., & Sędek, G. (1998). Uncontrollability as a source of cognitive exhaustion: Implications for helplessness and depression. In M. Kofta, G. Weary, & G. Sedek (Eds.), *Personal control in action: Cognitive and motivational mechanisms* (pp. 391–418). Plenum Press. https://doi.org/10.1007/978-1-4757-2901-6_16.

Lam, L. W., & Xu, A. J. (2019). Power imbalance and employee silence: The role of abusive leadership, power distance orientation, and perceived organisational politics. *Applied Psychology. An International Review, 68*(3), 513–546. https://doi.org/10.1111/apps.12170

Langenecker, S. A., & Nielson, K. A. (2003). Frontal recruitment during response inhibition in older adults replicated with fMRI. *NeuroImage, 20*(2), 1384–1392. https://doi.org/10.1016/S1053-8119(03)00372-0

Langenecker, S. A., Zubieta, J.-K., Young, E. A., Akil, H., & Nielson, K. A. (2007). A task to manipulate attentional load, set-shifting, and inhibitory control: Convergent validity and test-retest reliability of the parametric go/no-go test. *Journal of Clinical and Experimental Neuropsychology, 29*(8), 842–853. https://doi.org/10.1080/13803390601147611

Lenth, R. V. (2021). *Emmeans: Estimated marginal means, aka least-squares means*. https://cran.r-project.org/package=emmeans.

Liu, H., Han, F., Yuan, M., Lafferty, J., & Wasserman, L. (2012). The nonparanormal SKEPTIC. In *Proceedings of the 29th International Conference on Machine Learning, ICML 2012* (pp. 1415–1422). https://collaborate.princeton.edu/en/publications/the-nonparanormal-skeptic.

Liu, H., Lafferty, J., & Wasserman, L. (2009). The nonparanormal: Semiparametric estimation of high dimensional undirected graphs. *The Journal of Machine Learning Research, 10*, 2295–2328.

Maier, S. F., & Seligman, M. E. (1976). Learned helplessness: Theory and evidence. *Journal of Experimental Psychology: General, 105*(1), 3–46. https://doi.org/10.1037/0096-3445.105.1.3

Maier, S. F., & Seligman, M. E. P. (2016). Learned helplessness at fifty: Insights from neuroscience. *Psychological Review, 123*(4), 349–367. https://doi.org/10.1037/rev0000033

Morrison, E. W. (2023). Employee voice and silence: Taking stock a decade later. *Annual Review of Organizational Psychology and Organizational Behavior, 10*(1), 79–107. https://doi.org/10.1146/annurev-orgpsych-120920-054654

Morrison, E. W., See, K. E., & Pan, C. (2015). An approach-inhibition model of employee silence: The joint effects of personal sense of power and target openness. *Personnel Psychology, 68*(3), 547–580. https://doi.org/10.1111/peps.12087

Norman, D. A., & Shallice, T. (1986). Attention to action. In R. J. Davidson, G. E. Schwartz, & D. Shapiro (Eds.), *Volume 4. Consciousness and self-regulation: Advances in research and theory* (pp. 1–18). Springer US. https://doi.org/10.1007/978-1-4757-0629-1_1.

O'Donovan, R., De Brún, A., & McAuliffe, E. (2021). Healthcare professionals experience of psychological safety, voice, and silence. *Frontiers in Psychology, 12*. https://www.frontiersin.org/articles/10.3389/fpsyg.2021.626689.

Peterson, C., Maier, S. F., & Seligman, M. E. P. (1993). *Learned helplessness: A theory for the age of personal control*. Oxford University Press.

Pinder, C. C., & Harlos, K. P. (2001). Employee silence: Quiescence and acquiescence as responses to perceived injustice. In G. R. Ferris (Ed.), *20. Research in personnel and human resources management* (pp. 331–369). Emerald Group Publishing Limited. https://doi.org/10.1016/S0742-7301(01)20007-3.

Ric, F., & Scharnitzky, P. (2003). Effects of control deprivation on effort expenditure and accuracy performance. *European Journal of Social Psychology, 33*(1), 103–118. https://doi.org/10.1002/ejsp.134

Sambol, S., Suleyman, E., Scarfo, J., & Ball, M. (2023). A true reflection of executive functioning or a representation of task-specific variance? Re-evaluating the unity/diversity framework. *Acta Psychologica, 236*, Article 103934. https://doi.org/10.1016/j.actpsy.2023.103934

Schein, E. H. (2017). *Organizational culture and leadership*. Wiley.

Sędek, G., & Kofta, M. (1990). When cognitive exertion does not yield cognitive gain: Toward an informational explanation of learned helplessness. *Journal of Personality and Social Psychology, 58*, 729–743. https://doi.org/10.1037//0022-3514.58.4.729

Sherf, E. N., Parke, M. R., & Isaakyan, S. (2021). Distinguishing voice and silence at work: Unique relationships with perceived impact, psychological safety, and burnout. *Academy of Management Journal, 64*(1), 114–148. https://doi.org/10.5465/amj.2018.1428

Soral, W., Kofta, M., & Bukowski, M. (2021). Helplessness experience and intentional (un-)binding: Control deprivation disrupts the implicit sense of agency. *Journal of Experimental Psychology: General, 150*(2), 289–305. https://doi.org/10.1037/xge0000791

Sparrow, E. P., & Hunter, S. J. (2012). Foundations of executive function/dysfunction. In *Executive Function and Dysfunction: Identification, Assessment and Treatment*. Cambridge University Press. https://www.cambridge.org/core/books/executive-function-and-dysfunction/foundations-of-executive-functiondysfunction/623FEBDA8FE3D5341EDF1DF671D3D9A3.

Van Dyne, L., Ang, S., & Botero, I. C. (2003). Conceptualizing employee silence and employee voice as multidimensional constructs. *Journal of Management Studies, 40*(6), 1359–1392. https://doi.org/10.1111/1467-6486.00384

von Hecker, U., & Sędek, G. (1999). Uncontrollability, depression, and the construction of mental models. *Journal of Personality and Social Psychology, 77*(4), 833–850. https://doi.org/10.1037/0022-3514.77.4.833

von Hecker, U., Sędek, G., & McIntosh, D. N. (2000). Impaired systematic, higher order strategies in depression and helplessness: Testing implications of the cognitive exhaustion model. In U. von Hecker, S. Dutke, & G. Sedek (Eds.), *Generative mental processes and cognitive resources: Integrative research on adaptation and control* (pp. 245–275). Springer Netherlands. https://doi.org/10.1007/978-94-011-4373-8_10.

Votruba, K. L., & Langenecker, S. A. (2013). Factor structure, construct validity, and age- and education-based normative data for the parametric go/no-go test. *Journal of Clinical and Experimental Neuropsychology, 35*(2), 132–146. https://doi.org/10.1080/13803395.2012.758239

Whiteside, D. B., & Barclay, L. J. (2013). Echoes of silence: Employee silence as a mediator between overall justice and employee outcomes. *Journal of Business Ethics, 116*(2), 251–266. https://doi.org/10.1007/s10551-012-1467-3

Zelazo, P. D. (2015). Executive function: Reflection, iterative reprocessing, complexity, and the developing brain. *Developmental Review, 38*, 55–68. https://doi.org/10.1016/j.dr.2015.07.001

Zelazo, P. D., Craik, F. I. M., & Booth, L. (2004). Executive function across the life span. *Acta Psychologica, 115*(2), 167–183. https://doi.org/10.1016/j.actpsy.2003.12.005

Zhao, T., Liu, H., Roeder, K., Lafferty, J., & Wasserman, L. (2012). The huge package for high-dimensional undirected graph estimation in R. *Journal of Machine Learning Research, 13*, 1059–1062.

Zou, J., Zhu, X., Fu, X., Zong, X., Tang, J., Chi, C., & Jiang, J. (2025). The experiences of organizational silence among nurses: A qualitative meta-synthesis. *BMC Nursing, 24*(1), 31. https://doi.org/10.1186/s12912-024-02636-y